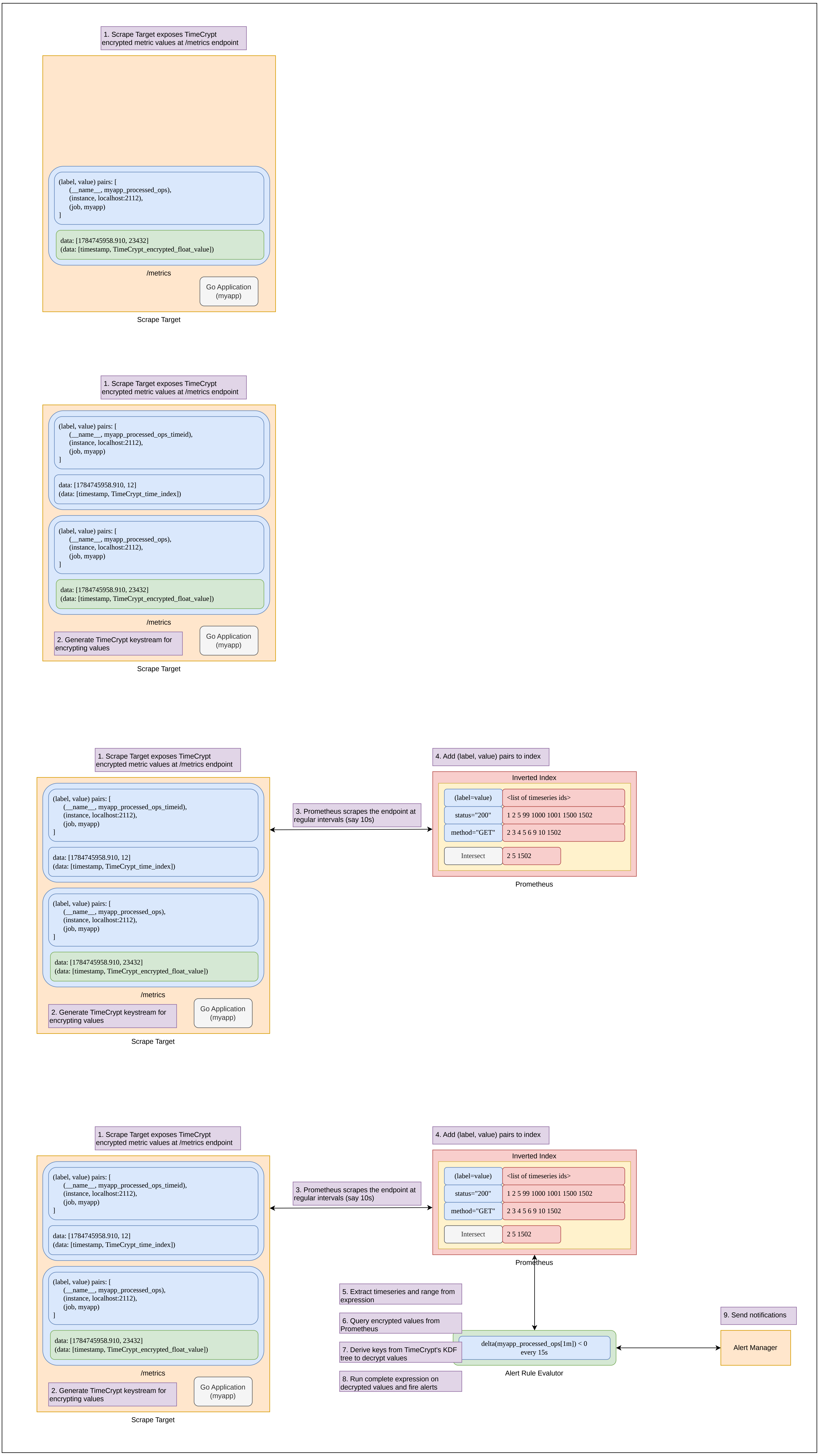
1. Scrape Target exposes TimeCrypt encrypted metric values at /metrics endpoint
(label, value) pairs: [
(__name__, myapp_processed_ops),
(instance, localhost:2112),
(job, myapp)
]
data: [1784745958.910, 23432]
(data: [timestamp, TimeCrypt_encrypted_float_value])
/metrics
Go Application (myapp)
Scrape Target
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(label, value) pairs: [
(__name__, myapp_processed_ops_timeid),
(instance, localhost:2112),
(job, myapp)
]
data: [1784745958.910, 12]
(data: [timestamp, TimeCrypt_time_index])
(label, value) pairs: [
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(instance, localhost:2112),
(job, myapp)
]
data: [1784745958.910, 23432]
(data: [timestamp, TimeCrypt_encrypted_float_value])
/metrics
2. Generate TimeCrypt keystream for encrypting values
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2. Generate TimeCrypt keystream for encrypting values
Go Application (myapp)
Scrape Target
3. Prometheus scrapes the endpoint at regular intervals (say 10s)
4. Add (label, value) pairs to index
Inverted Index
(label=value)
<list of timeseries ids>
status="200"
1 2 5 99 1000 1001 1500 1502
method="GET"
2 3 4 5 6 9 10 1502
Intersect
2 5 1502
Prometheus
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Prometheus
5. Extract timeseries and range from expression
6. Query encrypted values from Prometheus
7. Derive keys from TimeCrypt's KDF tree to decrypt values
8. Run complete expression on decrypted values and fire alerts
delta(myapp_processed_ops[1m]) < 0
every 15s
Alert Rule Evalutor
9. Send notifications
Alert Manager